UNIVERSITY OF GHANA

CPEN 211: DATABASE SYSTEM DESIGN

PROJECT REPORT

1. Introduction

Database systems play a critical role in modern e-commerce platforms by enabling efficient storage, retrieval, and management of large volumes of data. Online retail stores rely on databases to manage customers, product inventories, orders, and shipping details.This project focuses on designing a relational database system for an online retail store. The database is designed to store information about customers, products, orders, and shipping records. The system ensures efficient data organization, reduces redundancy, and supports reliable data retrieval.

The main objective of this project is to design an Entity Relationship (ER) model and translate it into a PostgreSQL database schema that supports the operations of an online retail store.

2. System Overview

The proposed system manages the core activities of an online retail store. Customers browse products, place orders, and receive shipped products.

The system performs the following functions:

- Stores customer information
- Maintains product inventory
- Records orders placed by customers
- Tracks shipping information for each order

When a customer places an order, the system records the order and the items included in it. Each order can contain multiple products. Shipping information is then linked to the order to track delivery status.

This database system ensures efficient organization and retrieval of retail data while maintaining data integrity.

3. Requirements Analysis

The system requires storage and management of the following key entities:

- **Customers**

Stores personal and contact information of customers.

- **Products**

Stores product information including price and stock quantity.

- **Orders**

Records purchase transactions made by customers.

- **Order Items**

Stores the individual products included in each order.

- **Shipping**

Tracks delivery information associated with orders.

The system must support the following relationships:

- A customer can place multiple orders
- An order can contain multiple products
- A product can appear in multiple orders
- Each order has shipping information

4. Entity Relationship (ER) Diagram

The ER diagram illustrates the entities, attributes, and relationships involved in the online retail system.

Entities

- Customer
- Product
- Order
- Order_Item
- Shipping

Relationships

Relationship	Alias	Description
Customer → Order	PLACES	One customer can place many orders
Order → Order_Item	CONTAINS	One order contains multiple items
Product → Order_Item	INCLUDES	Products appear in order items
Order → Shipping	HAS	Each order has shipping information

Cardinalities

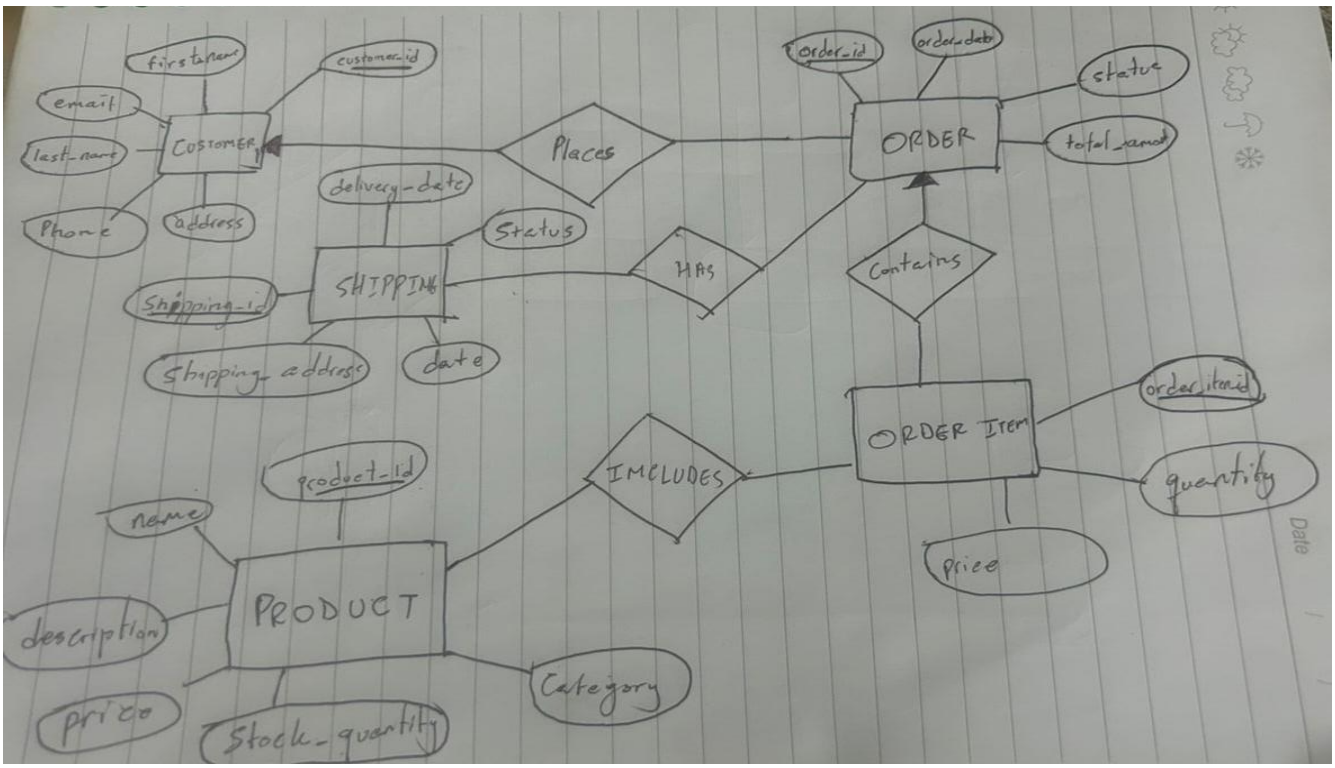
Customer (1) → (many) Orders

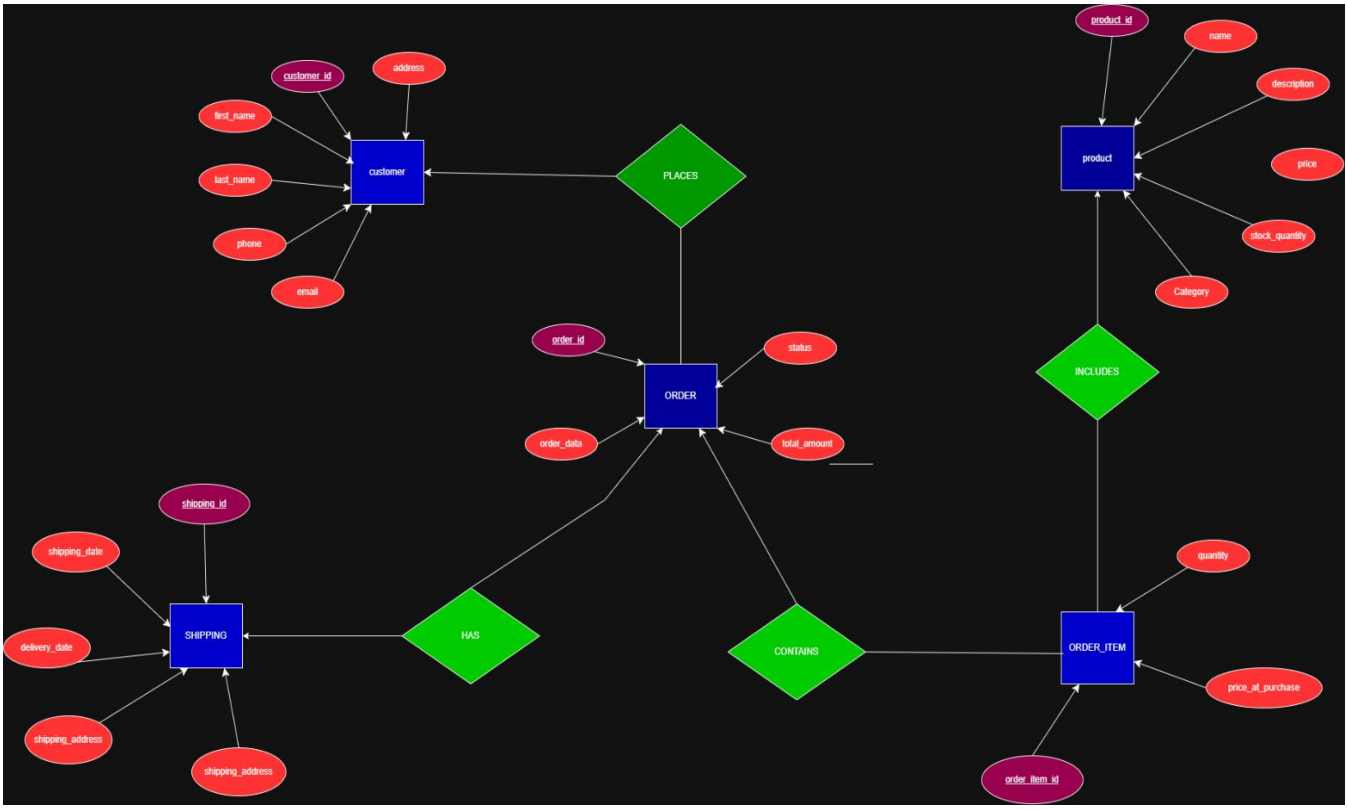
Order (1) → (many) Order_Items

Product (1) → (many) Order_Items

Order (1) → (1) Shipping

The ER diagram visually represents these entities and their interactions.


firstname
customer_id
email
last_name
CUSTOMER
Phone
address
Places
order_id
order_date
status
ORDER
total_amount
delivery_date
Status
SHIPPING
Shipping_id
Shipping_address
date
HAS
Contains
ORDER ITEM
order_item_id
quantity
price
IMCLUDES
product_id
name
PRODUCT
description
price
Stock_quantity
Category


customer_id
address
first_name
last_name
phone
email
customer
PLACES
product_id
name
description
product
price
stock_quantity
Category
order_id
status
ORDER
order_data
total_amount
INCLUDES
shipping_id
shipping_date
delivery_date
SHIPPING
shipping_address
shipping_address
HAS
CONTAINS
quantity
ORDER_ITEM
price_at_purchase
order_item_id

5. Database Schema Design

The ER model was translated into relational tables suitable for PostgreSQL.

Customers Table

Column	Type	Description
customer_id	SERIAL	Primary key
first_name	VARCHAR	Customer first name
last_name	VARCHAR	Customer last name
email	VARCHAR	Unique email
phone	VARCHAR	Contact number
address	TEXT	Customer address

Products Table

Column	Type
product_id	SERIAL
name	VARCHAR
description	TEXT
price	NUMERIC

Column	Type
stock_quantity	INTEGER

Orders Table

Column	Type
order_id	SERIAL
customer_id	INTEGER (FK)
order_date	TIMESTAMP
status	VARCHAR
total_amount	NUMERIC

Order_Items Table

Column	Type
order_item_id	SERIAL
order_id	INTEGER (FK)
product_id	INTEGER (FK)
quantity	INTEGER
price_at_purchase	NUMERIC

Shipping Table

Column	Type
shipping_id	SERIAL
order_id	INTEGER (FK)
shipping_address	TEXT
shipping_date	DATE
delivery_date	DATE
shipping_status	VARCHAR

The schema for the tables are present in the SQL file of the project.

6. Normalization

The database schema follows standard normalization principles.

- **First Normal Form (1NF)**

All attributes contain atomic values, meaning each column stores a single value.

Example:

Customer table stores one email per record.

- **Second Normal Form (2NF)**

All non-key attributes depend on the entire primary key.

Example:

In the Order_Items table, quantity and price depend on the order item.

- **Third Normal Form (3NF)**

There are no transitive dependencies.

Example:

Customer information is stored only in the Customers table, avoiding duplication across orders.

This ensures reduced redundancy and improved data integrity.

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7. Implementation in PostgreSQL

The following SQL scripts were used to create the database tables.

Example:

```
CREATE TABLE customers (
customer_id SERIAL PRIMARY KEY,
first_name VARCHAR(50),
```

```
last_name VARCHAR(50),
email VARCHAR(100) UNIQUE,
phone VARCHAR(20),
address TEXT
);
```

Foreign key relationships ensure referential integrity between tables.

Additional constraints such as NOT NULL, UNIQUE, and CHECK help maintain valid data.

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8. Sample Queries

Example queries demonstrating system usage include:

- **Retrieve all customer orders**

```
SELECT customers.first_name, orders.order_id
FROM customers
JOIN orders
ON customers.customer_id = orders.customer_id;
```

- **List all products**

```
SELECT * FROM products;
```

Track shipping status

```
SELECT order_id, shipping_status
FROM shipping;
```

These queries illustrate how the database supports typical operations of an online retail store.

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9. System Demonstration

During the demonstration, the database system was tested using PostgreSQL.

The following operations were demonstrated:

- Creating database tables
- Inserting customer and product records
- Creating orders
- Linking products to orders
- Retrieving shipping status

The demonstration confirmed that the system successfully manages customer, product, and order data.

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10. Conclusion

This project demonstrated the design and implementation of a relational database for an online retail store.

The ER model was used to identify entities, relationships, and attributes. The model was then translated into a PostgreSQL relational schema with appropriate primary and foreign key constraints.

Normalization techniques were applied to reduce redundancy and ensure efficient data storage.

The resulting system supports the management of customers, products, orders, and shipping records, providing a scalable and reliable database solution for an online retail environment.